Valuation Scheme
25900364

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACECT200, MG3VACECC200 - GREEN ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A
Multiple Choice Answers

1. a	**19. a**
2. b	**20. b**
3. b	**21. a**
4. a	**22. a**
5. b	**23. c**
6. b	**24. d**
7. a	**25. c**
8. b	**26. c**
9. c	**27. b**
10. b	**28. d**
11. c	**29. b**
12. c	**30. b**
13. d	**31. b**
14. b	**32. b**
15. a	**33. a**
16. a	**34. d**
17. b	**35. b**
18. c	

Part B
Short Essay Questions (5*3=15)

36. Differentiate hazardous vs non-hazardous compounds in e-waste with examples.

Definition of hazardous and non-hazardous compounds-3 marks

Examples-2 marks

E-waste contains a variety of materials that can be classified as hazardous and non-hazardous based on their chemical composition and impact on the environment and human health.

Hazardous compounds:

These are substances that pose serious risks due to their toxic, corrosive, or flammable nature. Common examples include:

- Lead (Pb): Found in solder, cathode ray tubes (CRTs), and circuit boards; can cause neurological damage.
- Mercury (Hg): Present in switches, fluorescent lamps, and flat screens; causes kidney and brain damage.

- Cadmium (Cd): Used in batteries and semiconductors; a known carcinogen.

- Brominated flame retardants (BFRs): Present in plastic casings and cables; persistent organic pollutants (POPs).

Non-hazardous compounds:

These materials are generally stable and pose minimal direct health risks. Examples include:
- Plastics (PVC, ABS): Used for insulation and casings.
- Glass: Found in monitors and screens.
- Metals like iron, aluminum, and copper: Recyclable and valuable for recovery.

Proper segregation of hazardous and non-hazardous components is crucial for safe handling, recycling, and environmental protection.

37. Explain bioleaching and its role in gold recovery from e-waste.

Definition of bioleaching-3 marks

Recovery of gold-2 marks

Bioleaching is an eco-friendly technique that uses microorganisms to extract metals from solid waste materials such as ores or electronic waste. It is an alternative to conventional chemical leaching methods.

Principle: Certain bacteria (like Acidithiobacillus ferrooxidans and Thiobacillus thiooxidans) produce organic acids or oxidizing agents that dissolve metals from the waste material.

Process in e-waste:In the context of printed circuit boards (PCBs) and other electronic scraps, bioleaching helps to:
1. Break down the base metal matrix using bacterial activity.
2. Expose and release precious metals such as gold, silver, and copper.
3. Reduce the need for harmful chemicals like cyanide used in traditional leaching.

Advantages:
- Environmentally friendly and cost-effective.
- Operates under mild conditions (ambient temperature and pressure).
- Produces less chemical waste.

Limitations:
- Slower process rate compared to chemical leaching.
- Requires controlled environmental conditions (pH, temperature, nutrient supply).

Thus, bioleaching offers a sustainable approach for gold recovery from e-waste with minimal ecological impact.

38. Discuss one major limitation of paper-based dielectric materials in electronic applications.

Definition of paper based dielectric material-2 marks

Limitations (Any 3)- 3 marks

Paper-based dielectrics are lightweight, biodegradable, and low-cost materials that have attracted attention in flexible and sustainable electronics. However, they have significant limitations that restrict their large-scale application.

The major limitation of paper-based dielectric materials is their poor moisture resistance and low dielectric stability under humid or high-temperature conditions. Paper is inherently hygroscopic — it absorbs moisture from the environment, leading to:
- Reduced dielectric strength and insulation resistance.
- Swelling and dimensional instability.
- Increased dielectric losses and short circuits.

39. Evaluate the risks of applying air classification before shredding during e-waste processing.

Air classification -2 marks

Limitations-3 marks

Air classification is a technique used to separate materials based on differences in density and aerodynamic properties. However, applying air classification before shredding can pose several operational and environmental risks.
1. **Ineffective separation**: Large and intact e-waste components such as monitors, CPUs, or cables have irregular shapes, which reduces the efficiency of air-based separation before size

reduction.

2. **Equipment damage and blockages**: Bulky items can obstruct air ducts and classifiers, leading to mechanical failures or unplanned downtime.

3**. Dust and contamination risks**: Air movement can disperse fine dust containing toxic substances like lead, cadmium, and brominated flame retardants, posing health hazards to workers and contaminating the workspace.

4**. Reduced overall efficiency**: Pre-classification without shredding prevents accurate segregation of materials by weight or density, decreasing the recovery rate of valuable metals.

Therefore, air classification should be applied after controlled shredding under enclosed conditions with proper dust collection systems to ensure safety and process efficiency.

40. List and explain the computer components that can be reused from discarded PCs.

Any 5 components-5 marks

Discarded personal computers (PCs) consist of many reusable components that can be tested, refurbished, and used again, reducing e-waste generation and resource consumption.

Reusable components include:

1. Memory modules (RAM): Can be reused in compatible systems after testing for stability and capacity.
2. Storage devices (HDD/SSD):Suitable for reuse after secure data wiping and performance verification.
3. Power supply units (SMPS):Reusable after electrical testing and dust removal.
4. Cooling fans and heat sinks:Often in good condition and can be cleaned and reused.
5. Cables and connectors: Useful for other devices if insulation and terminals are intact.
6. Computer cases and enclosures:Can be reused for assembling new systems.
7. Monitors and peripheral devices (keyboard, mouse): If functional, can be reused in other setups or educational labs.

Benefits:

Reuse extends the product life cycle, minimizes electronic waste, reduces energy consumption in manufacturing new parts, and supports sustainable electronic practices.